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$$\begin{aligned}& \int \frac{5dx}{(4x^2 + 1)^2} \\& \text{Stel } \begin{cases} t = 2x \\ dt = 2dx \end{cases} \\& \Rightarrow \frac{5}{2} \int \frac{dt}{(t^2 + 1)^2} \\& = \frac{5}{2} \cdot \frac{1}{2} \left(\text{Bgtan}(t) + \frac{t^2}{t^2 + 1} \right) + C \\& = \frac{5}{4} \text{Bgtan}(2x) + \frac{10x}{4((2x)^2 + 1)} + C\end{aligned}$$

References